

# MakerWorld Quick Start Guide

## Braille Cylinder STL Generator — Parametric Model Maker edition

This guide walks you from "I want a braille label" to a downloadable STL pair using the single-file build in [makerworld/Braille Cylinder STL Generator MakerWorld.scad](#) on [MakerWorld's](#) Parametric Model Maker.

A PDF version of this guide is available: [MakerWorld Quick Start Guide.pdf](#).

## 1. What to Include

Braille takes far more space than print — you cannot include everything, so prioritize. The core question, per the Braille Authority of North America (BANA): "*Can someone identify me and contact me with just this information?*" If yes, you have enough.

- Each row holds **13 braille cells of text** by default, and the generator offers **4 lines** (Line\_1–Line\_4).
- BANA's typical four-line business card layout: **name, organization, phone number, e-mail address**. Job titles, mailing addresses, fax numbers, and websites are usually the first things to cut.
- If a name will not fit on one 13-cell line: drop capital indicators, drop a middle initial, use a first initial, or continue the name onto the next line.
- Organization names can be abbreviated (BANA's examples: lib for "library", amer for "American", nat for "National") or omitted entirely if the e-mail address already contains them.

Source: BANA *Business Cards Fact Sheet* (approved March 2024) — see Resources below.

## 2. Translate Your Text

MakerWorld's customizer cannot translate English for you. Paste **pre-translated Unicode braille**, one line per Line\_N field.

1. Open a braille translator such as <https://www.branah.com/braille-translator>.
2. Type your text in **lowercase** (each capital letter adds an indicator cell — braille readers understand omitting capitals is standard practice for space-limited labels).
3. Copy the **Unicode braille** output — the dot characters such as ⠠⠠⠠⠠⠠⠠ — not the ASCII/BRF letters.
4. Translate each line separately so you control where lines break.

Uncontracted (Grade 1) UEB is clearest for names and contact information.

## 3. Formatting

### Phone numbers

BANA guidance: braille a phone number as #123.456.7890 — omit parentheses and convert hyphens to periods. In Unified English Braille a period inside a number keeps *numeric mode* active, so only **one** number sign (⠠) is needed for the whole number. For example, 206.616.7678 translates to:

⠠⠠⠠⠠⠠⠠⠠⠠⠠⠠⠠⠠⠠⠠ — exactly **13 cells**, which fits the default row.

Some online translators repeat the number sign after each period. That output is non-standard and wastes cells — if your translation shows more than one ⠠ in a phone number, it likely will not fit.

**Splitting a long number across two rows:** divide after a period and begin the next row with a fresh number sign:

Row 1: ⠠⠠⠠⠠⠠⠠⠠⠠⠠⠠⠠⠠⠠⠠ (206.616.)

Row 2: ⠠⠠⠠⠠⠠⠠⠠⠠⠠⠠⠠⠠⠠⠠ (7678)

## E-mail and web addresses

Braille them per the Rules of UEB. When an address must be divided over two lines, divide after punctuation (the @ sign, a period, or a hyphen) and begin the second line in cell 1. A dot-5 line continuation indicator at the end of the first line is preferred; omit it only as a last resort. Tip: omit https:// or www. if the domain is clear without it.

## 4. Using the Customizer

1. Go to MakerWorld → **Create** → **Parametric Model Maker** and upload **only** Braille\_Cylinder\_STL\_Generator\_MakerWorld.scad.
2. In the parameter panel:
  - Paste your Unicode braille into Line\_1, Line\_2, Line\_3, Line\_4 (leave unused lines empty). Do **not** type plain English.
  - paper\_thickness\_preset: keep 0.4mm (default) for typical card stock, 0.3mm for thinner paper, or Custom to use your own slider values. Presets control dot and spacing dimensions; they never change the grid\_columns / grid\_rows capacity sliders.
  - dot\_shape: Cone (default) or Rounded.
  - indicators: On adds a triangle/rectangle marker at each row's start and end for plate alignment. This widens the grid by 2 marker cells — your 13-cell text capacity is unchanged.
3. Generate the **Embossing Plate** (plate\_type = Embossing Plate), render, and download the STL.
4. Switch plate\_type to **Counter Plate** — change nothing else — and download again. The two plates form a matching pair: paper goes between them, and pressing the embossing plate's dots into the counter plate's recesses forms readable braille.

## 5. Cylinder Sizing

- **Measure the diameter** of the container you are labeling: wrap a string around it, measure the string, and divide by  $\pi$  (3.14159). Enter it as cylinder\_diameter.
- **Height:** measure the flat area where the braille will sit and leave a margin at the top and bottom.
- **Capacity math:** cells are spaced 6.5 mm apart, so the default 13 text cells + 2 indicator cells span about 91 mm. The default 30.8 mm-diameter cylinder (~96.8 mm circumference) fits that with room left for the seam gap. A larger diameter adds margin around the seam rather than more cells — raise grid\_columns yourself if you want longer rows on a bigger cylinder. With indicators Off, up to 15 text cells fit the default cylinder.
- Print cylinders standing upright for the best dot quality, and use seam\_offset\_degrees to rotate the seam away from your text.

## 6. Troubleshooting

### INVALID CHARACTERS

Red 3D text reading INVALID CHARACTERS means a Line\_N field contains something other than Unicode braille (typed English, ASCII braille, or stray spaces from another alphabet). Re-translate and copy the Unicode braille output.

### TEXT TOO LONG: 16/13

A line exceeds the row capacity. The warning shows the counts — longest line vs. capacity — because MakerWorld's preview cannot show console output. Fixes, in order of preference:

1. **Shorten or split** the line across rows (see the phone-number splitting pattern above).
2. **Raise grid\_columns** (Braille Grid Layout section) if your cylinder is large enough.

3. **Set text\_limit\_check to Off** (Text Input section) to bypass the check entirely: every pasted cell renders and no warning appears, but rows longer than the capacity may crowd the seam gap.

## 7. Examples

Business-card examples take the content decisions from BANA's Fact Sheet (BANA's published braille is Grade 2; translate your own text as Grade 1).

| Scenario                               | Lines to translate and paste                                   | Notes                                                            |
|----------------------------------------|----------------------------------------------------------------|------------------------------------------------------------------|
| Name + e-mail (BANA Example 1)         | harry potter / harry@hogwarts.edu                              | Organization omitted — it already appears in the e-mail address. |
| Long name (BANA Example 4)             | liesel a. / schimmelfennig / l.schimmelfennig@usace.army       | Name continues onto the second line; phone omitted.              |
| Nickname + two phones (BANA Example 7) | fran rikard / albuquerque ac / c 505.312.4224 / f 505.312.4225 | Nickname saves cells; c/f prefixes tag cell and fax.             |
| Spice jar (~55 mm diameter)            | cinnamon                                                       | One row; set cylinder_diameter = 55.                             |
| Medicine bottle (~40 mm)               | amoxicillin / 500mg                                            | "amoxicillin" is 11 cells — within the 13-cell row.              |
| Water bottle tag (~75 mm)              | j. smith / 555.867.5309                                        | Initials save space; the phone number fits one row.              |

## 8. Resources

- [BANA Position Statements and Fact Sheets](#)
- [BANA Business Cards Fact Sheet \(PDF, approved March 2024\)](#)
- [BANA Size and Spacing of Braille Characters](#)
- [The Rules of Unified English Braille \(ICEB\)](#)
- [This project on GitHub](#)
- [Web app companion project](#) — translates English for you in the browser
- For complex cases (international phone numbers, multiple languages, credentials), consider working with a **UEB-certified transcriber**.